

Aditya Kulkarni

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Raleigh, NC (open to relocating to New York)

PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) with production experience building and shipping AI agent systems in Python and TypeScript. At Isomer AI, built Temporal.io workflow orchestration and multi-model AI agent tooling across Claude, GPT-4o, and Gemini using MCP -- 394 commits, 7 production deployments on a live product. Independently built AlgoPulse, a fault-tolerant agentic platform processing 2,000 async inference requests/second, and Glance AI, a compliance-ready RAG pipeline for fintech and healthcare. Case studies: kulkarniaditya.com

CORE COMPETENCIES

AI Agent Development

Conversational AI Systems

Python & TypeScript

Async Programming at Scale

Production LLM Systems

RAG Pipeline Design

Multi-model LLM Evaluation

Temporal.io & Kafka

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern

- Built and shipped multi-model AI agent tooling across 3 LLM providers (Anthropic Claude, OpenAI GPT-4o, Google Gemini) and 8+ models using MCP, with **Temporal.io workflow orchestration** for durable, fault-tolerant agent execution -- 394 commits, 7 production deployments
- Developed 10+ backend features including role-based access control, real-time event pipelines (Pusher), and document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 storage) in TypeScript / Node.js / PostgreSQL / Redis
- Integrated Sentry observability and ran evaluations across 8+ model variants to tune agent reliability and output quality in production

NC State University Apr 2025 - Present

Graduate Research Assistant

- Designed a cloud-based genomic data platform with JBrowse 2, serving 500GB of datasets to 100+ researchers worldwide on Google Cloud Run
- Applied hyperparameter tuning and time-series ML analysis, improving model accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows including a health claim automation tool using OCR, NLP, and ML pipelines; improved claim accuracy by 25%
- Automated deployments on AWS and GCP; integrated Grafana monitoring; supported model validation and red-teaming

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Led end-to-end SDLC for a support ticket analysis system; built Hadoop + Spark batch pipelines and Kafka streaming -- reduced ticket-processing latency by 20%

PROJECTS

AlgoPulse -- Agentic AI Platform Independent, 2025

Architected a fault-tolerant, event-driven AI orchestration platform in Python and TypeScript. Identified API-polling bottleneck under load; re-architected to Supabase Edge Functions streaming AI reasoning loops directly to clients. Temporal.io manages durable async workflow state; Kafka handles worker coordination. Scaled to **2,000 async inference requests/second** with zero message loss -- same throughput profile as enterprise conversational AI at scale.

Python, TypeScript, Node.js, Supabase Edge Functions, Temporal.io, Kafka

Glance AI -- Privacy-First RAG Pipeline Independent, 2025

Built a compliance-ready RAG system for fintech and healthcare with per-tenant data isolation at the vector retrieval layer. Prevents cross-tenant context leakage and LLM hallucinations; deterministic output architecture for regulated environments.

Python, LLMs, Vector Databases, RAG

PerCent -- GenAI Finance App HackNC24 Top 10 / 100+ teams

Built a conversational AI chatbot using Qwen-14B-Chat and RAG for financial domain reasoning; reduced manual research time by 30%. Shipped complete MVP in 48 hours -- goal tracking, expense logging, AI chatbot -- Top 10 of 100+ teams.

Python, Selenium, Hugging Face, Qwen-14B-Chat, RAG

CryptoKnight -- Trustless AI Trading Agent HackNC State, 2025

Built a trustless autonomous AI agent using ERC-4337 Account Abstraction with Session Keys enforcing on-chain policies (trade limits, allowance cap, expiry, revocation). Agent acts fully autonomously within cryptographically enforced constraints -- no private keys exposed.

ERC-4337, Python agent, Next.js 16, viem, wagmi

EDUCATION

Master of Science, Computer Science North Carolina State University

Aug 2024 - May 2026

Courses: Artificial Intelligence, Operating Systems, Data Science, Algorithms, Software Engineering

B.Tech, Computer Science Mumbai University

Aug 2020 - May 2024

CGPA: 9.32/10 | Courses: Machine Learning, Big Data Infrastructure, Blockchain, Algorithms

SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C

AI/ML: LLM APIs (Claude, GPT-4o, Gemini), MCP, LangChain, RAG, Hugging Face, TensorFlow, PyTorch, Prompt Engineering

Agent Infra: Temporal.io, Kafka, Async Programming, Distributed Systems, Supabase Edge Functions

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, CI/CD, Grafana, Sentry

Frameworks: Node.js, React, Express, Prisma, PostgreSQL, Redis, Auth0